📊Multi-Channel Marketing Campaign Analysis

1. Introduction

In this project, I performed a detailed analysis of a **multi-channel marketing campaign** to identify the most effective campaign and channel combination for driving **new customer growth**.

The analysis focuses on understanding how different campaigns (**A vs. B**) and channels (**Email, Instagram, Web Banner**) perform across key business metrics such as **conversion rate**, **revenue**, and **customer acquisition**. The goal is to recommend a **data-driven strategy** aligned with specific business objectives.

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2. Problem Statement

The core business question is:

👉 **Which campaign and channel combination should be prioritized to maximize new customer growth?**

Additionally, the analysis explores:

- Which **campaign** performs better overall?
- Which **channel** drives the highest conversions and revenue?
- Whether there is a **trade-off** between customer volume and customer value.

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3. Dataset Description

The dataset contains interaction-level data with the following columns:

- **Interaction ID:** Unique identifier for each interaction.
- **Campaign Type:** A (Informational), B (Promotional).
- **Channel:** Email, Instagram, Web Banner.
- **Customer Type:** New or Existing.
- **Converted:** 1 (Yes), 0 (No).
- **Time on Site:** Engagement metric measured in seconds.
- **Sales ($):** Revenue generated.

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4. Data Preparation

Before performing the analysis, the dataset was verified to ensure:

- No **missing or null values** in key columns.
- Correct **data types** for numerical and categorical variables.
- Clean and consistent **category labels**.

The data was already cleaned, so no major preprocessing steps were required.

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5. Key Metrics Defined

To evaluate performance, the following metrics were used:

5.1 Conversion Rate

$$\text{Conversion Rate} = \frac{\text{Number of Conversions}}{\text{Total Users}}$$
This metric measures how effectively a campaign converts users into customers.

5.2 Revenue

$$\text{Revenue} = \sum \text{Sales}$$
Indicates the **total business impact**.

5.3 New Customer Revenue

Revenue generated specifically from **new customers**. This aligns directly with the primary business goal.

5.4 Customer Acquisition

The total count of **new customers** who converted. Used to measure growth scale.

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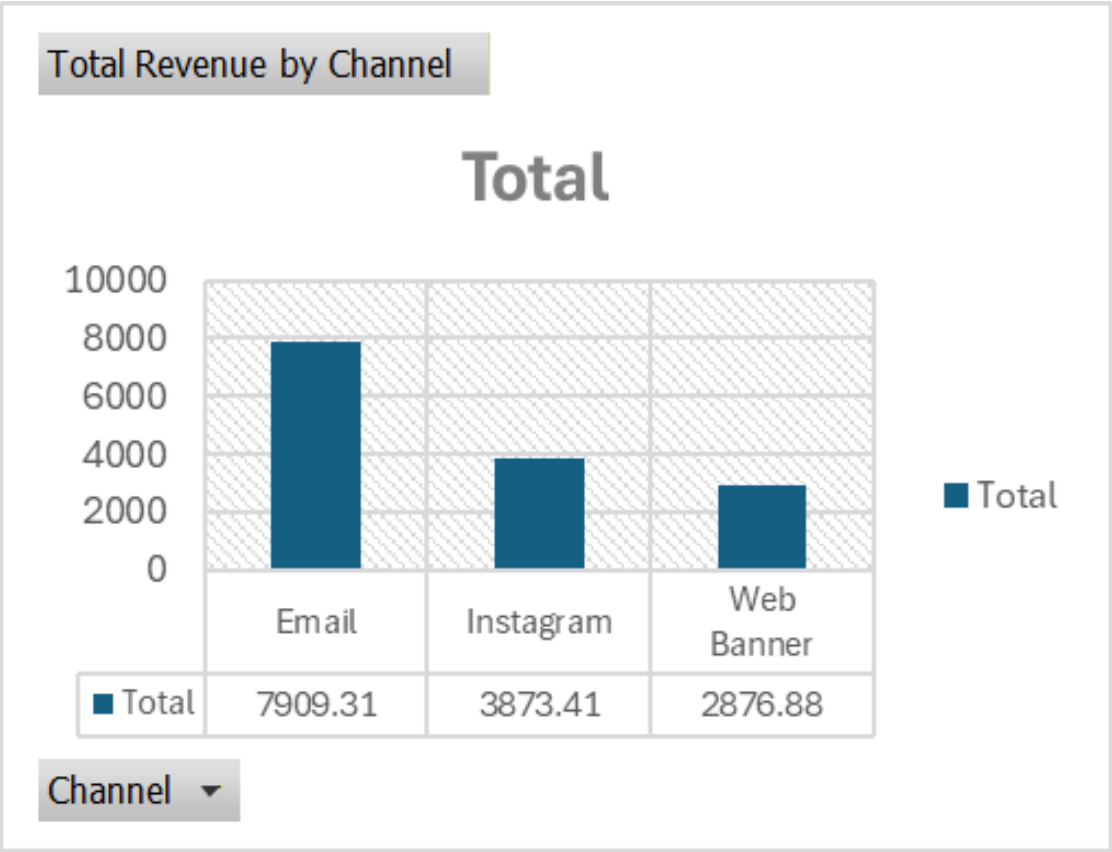
6. Exploratory Data Analysis

6.1 Channel Performance

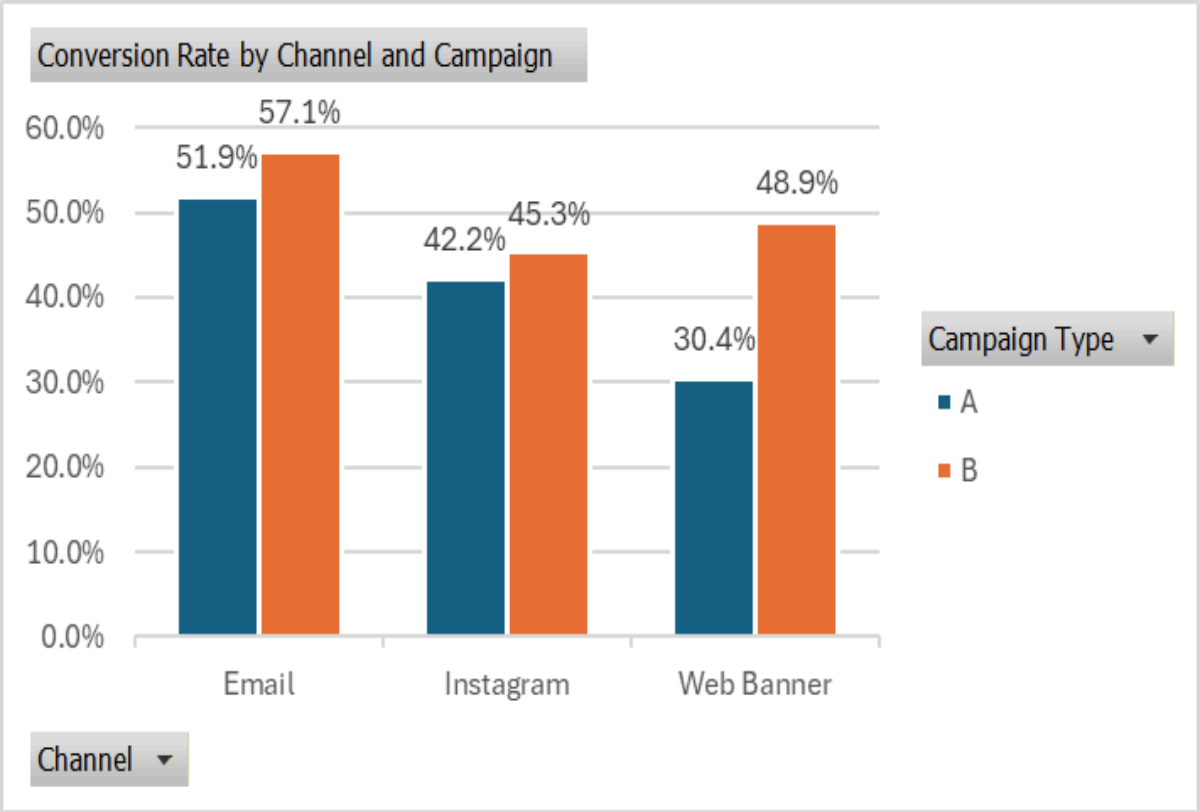
- **Email** consistently performs best across both campaigns.
- **Instagram** shows moderate engagement but lower conversion.
- **Web Banner** performs weakest in both conversion and revenue.



Insight: Email drives the highest revenue across all channels.

6.2 Campaign Performance

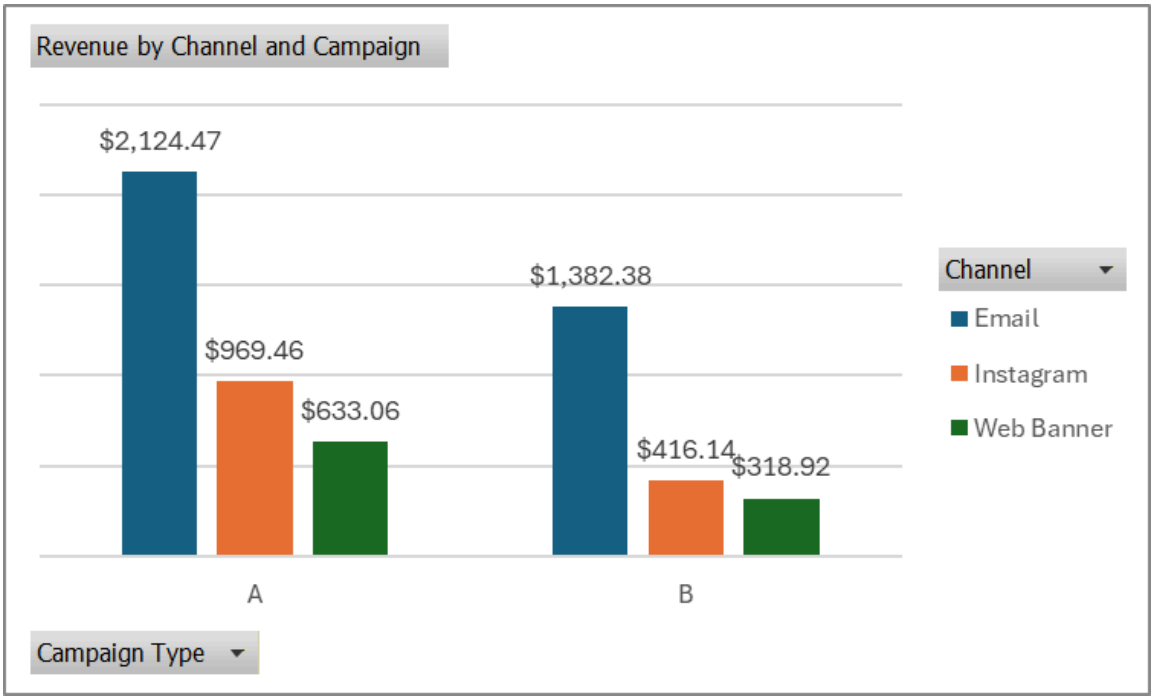
- **Campaign B** shows higher **conversion rates** across all channels.
- **Campaign A** generates higher **revenue per customer**.



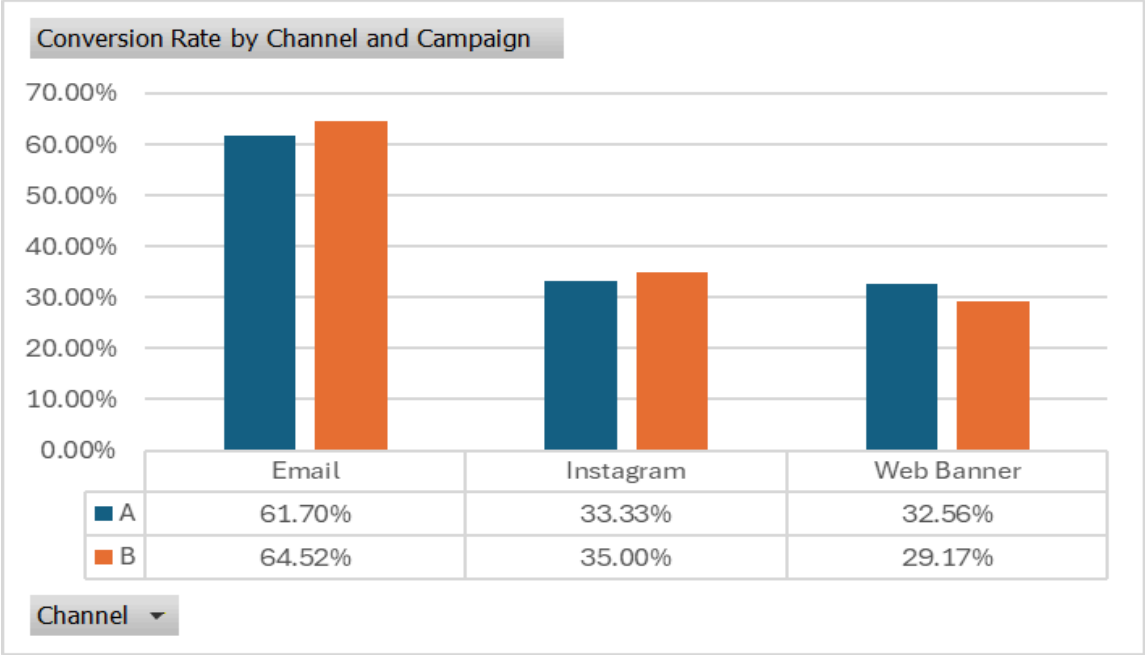
Insight: Campaign B drives the higher Conversion across all channels.

6.3 New Customer Analysis

- **Email + Campaign A** generates higher revenue per new customer.



- **Email + Campaign B** generates more total conversions.



	Email	Instagram	Web Banner
A	61.70%	33.33%	32.56%
B	64.52%	35.00%	29.17%

This indicates that Email is the most effective channel, while Web Banners underperform and may require reduced investment.

7. Key Insight (Critical Finding)

The analysis reveals a clear **trade-off**:

- **Campaign A:** Higher revenue per customer (**Value**).
- **Campaign B:** Higher conversion rate (**Volume**).

This indicates that no single campaign is universally "best"—the choice depends entirely on the current **business goal**.

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8. Business Interpretation

- **Campaign B** is more effective for **scaling customer acquisition**.
- **Campaign A** is more effective for **maximizing revenue per customer**.

This distinction is critical for making strategic budgetary decisions.

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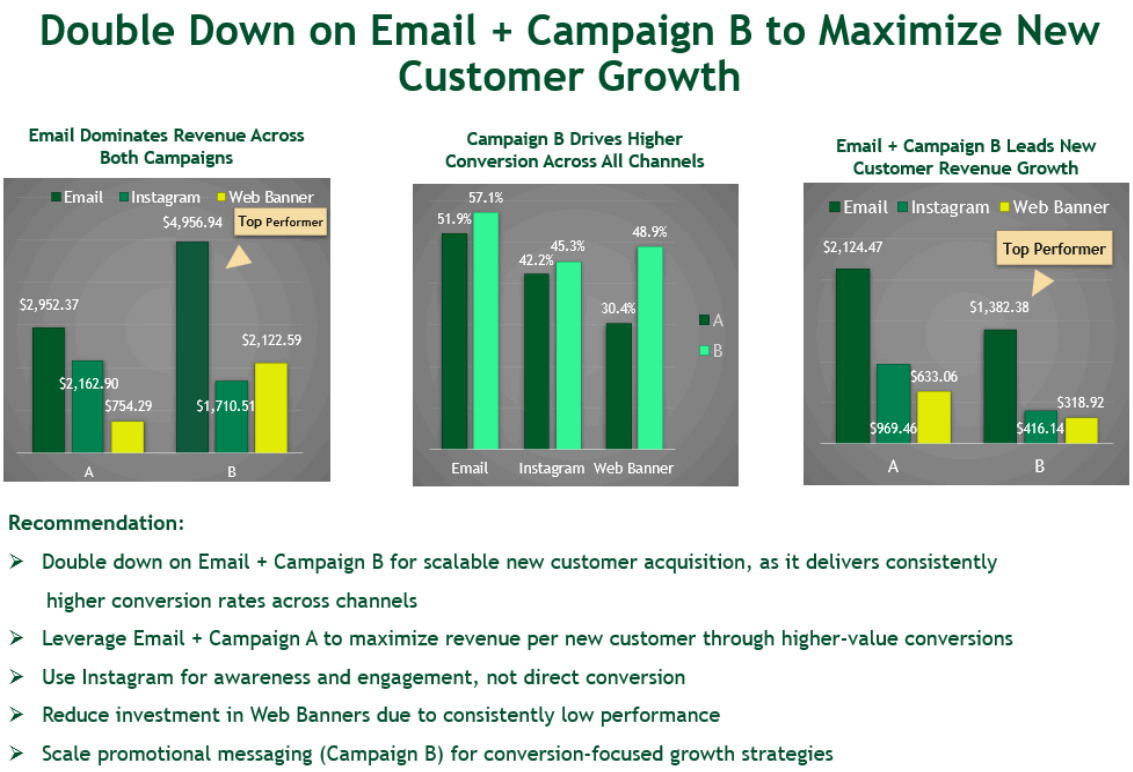
9. Recommendation

Based on the analysis, the following actions are recommended:

- **Double down** on **Email + Campaign B** for scalable new customer acquisition.
- Use **Email + Campaign A** strategically to maximize revenue per customer.
- Use **Instagram** primarily for **awareness and engagement**.
- **Reduce investment** in **Web Banner** due to low performance.



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10. Validation Strategy (A/B Testing)

To validate these recommendations, we will split users into two groups:

- **Group A:** Email + Campaign A
- **Group B:** Email + Campaign B

We will track **Conversion Rate**, **New Customers Acquired**, **Revenue**, and **Customer Retention** over time to identify the superior long-term strategy.

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11. Measuring Success

Success will be evaluated based on:

- Increase in **new customer acquisition**.
- Improvement in **overall conversion rate**.
- Growth in **total revenue**.
- **Quality of customers** (retention and repeat purchases).

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12. Conclusion

This analysis demonstrates that effective decision-making in data analytics requires understanding both **performance metrics** and **business objectives**. Rather than selecting a single “best” campaign, the optimal strategy involves balancing **scale (Campaign B)** and **value (Campaign A)**.

Would you like me to help you draft the specific **A/B test parameters** or create a **summary table** for a presentation?